Lesson plan · for the teacher

Networks: getting the data home

IoT · Level 3 · Lesson 4

Time
40 min

Materials
Paper notes and a “gateway” sign; An open space: the playground or a long corridor; Chalk or markers to draw the school plan; Optional: two phones, to see how far a Bluetooth connection reaches as pupils walk apart

Learning goals

- Compare Bluetooth, Wi-Fi, mobile networks and LoRa by range, power, data and cost.
- Choose a network for a school sensor from its distance, power and data needs.
- Explain how gateways and relays (a mesh) carry messages beyond one radio's reach.

Activity: The playground radio game

1. Warm-up (5 min): Draw the school and a water tank far from the office. Ask: the sensor knows the tank is nearly empty, so how can that reading reach the caretaker's phone?
2. Teach (8 min): Ask the four questions (how far, what power, how much data, what cost). Compare Bluetooth LE, Wi-Fi, a mobile network and LoRa in a table on the board, with one more row for a memory card collected weekly.
3. Play (15 min): In the playground, one pupil is the gateway by the door; the others spread out as sensors. A note may pass only to someone within arm's reach. Start notes at the farthest pupils and count the hops. Who is cut off? Add a relay pupil, then take one pupil away and find another route.
4. Choose (8 min): Back in class, groups pick one real sensor at school (the tank, the garden or a classroom) and choose a network, giving reasons for distance, power, data and cost.
5. Wrap-up (4 min): Ask: what happened when a relay pupil left, and how did your network find another route?

Choose a safe, open space and agree a signal to stop. Pupils walk, never run, and pass notes only. A pupil who can't move about easily can be the gateway.

Discussion

- Which school sensor would you connect first, and how?
- What happens to the data if the gateway loses power?
- When would a memory card collected weekly be enough?

Exit ticket

- A soil sensor 2 km from school, on batteries, sends 10 bytes an hour. Which network, and why?

Try it at school tomorrow
The playground radio game: In the playground, one pupil is the gateway by the classroom door. The others spread out as sensors; a folded note may pass only to someone within arm's reach. Start a note at the farthest pupil and count the hops. Who is cut off? Add a relay pupil to bridge the gap, then take one pupil away and find another route. Back in class, draw the school on the board and choose a network for one real sensor, with reasons: distance, power, data and cost.

Practice questions · for you

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1 **Kofi's soil sensor at the far end of the farm can't reach the gateway, but a sensor halfway between them can reach both. What is the cheapest fix?**

- ◯ Make the far sensor send more often
- ◯ Give the far sensor a bigger battery
- ◯ Let the halfway sensor pass the messages on

2 **Match each school need to the network that fits it best.**

A pupil's step counter that syncs to a phone	• • • •	Wi-Fi
A CO_2 sensor in the staffroom, beside the router	• • • •	A mobile network (SIM and plan)
A tracker on the school bus, all over town	• • • •	Bluetooth Low Energy
Soil sensors on a farm 3 km from the school	• • • •	A memory card collected weekly
A week of classroom temperatures, read on Fridays	• • • •	LoRa, with a gateway at school

3 **A LoRa sensor can send a photo of the school garden every hour.**

- ◯ True
- ◯ False

4 **In the mesh lab, which of these are true?**

- ☐ Every extra hop saves battery
- ☐ Raising the range to 500 m cuts the most hops from 5 to 3
- ☐ At 240 m, six sensors are cut off
- ☐ A relay only passes messages on; it isn't a sensor
- ☐ If any one sensor breaks, the whole network stops

5 **A tank sensor sends a 4-byte reading every 15 minutes.**

That's _____ messages a day, and _____ bytes a day.

Word bank: 384 · 1,440 · 96 · 24

✎ **Sketch your school: where would one useful sensor go, how far is it from power and Wi-Fi, and which network would you choose?**

Check your answers

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1 **Let the halfway sensor pass the messages on**

Range doesn't grow with more sending or a bigger battery.

2 **A pupil's step counter that syncs to a phone — Bluetooth Low Energy**
A CO_2 sensor in the staffroom, beside the router — Wi-Fi
A tracker on the school bus, all over town — A mobile network (SIM and plan)
Soil sensors on a farm 3 km from the school — LoRa, with a gateway at school
A week of classroom temperatures, read on Fridays — A memory card collected weekly

Match the distance, power and data: the phone is close, the router is in the room, the bus moves through mobile coverage, the farm is kilometres away with tiny readings, and a weekly report needs no radio at all.

3 **False**

LoRa carries only a few dozen bytes at a time, and a phone photo is a few million bytes: tens of thousands of messages, far more than LoRa allows.

4 **At 240 m, six sensors are cut off; A relay only passes messages on; it isn't a sensor; Raising the range to 500 m cuts the most hops from 5 to 3**

At 240 m, six sensors have no chain of neighbours in range to the gateway; relays bridge the gaps without measuring anything.

5 **That's 96 messages a day, and 384 bytes a day.**

Four readings an hour × 24 hours = 96 messages, and 96 × 4 = 384 bytes, about a fifth of a page of text.